TOPFAITH UNIVERSITY
MKPATAK, AKWA IBOM STATE, NIGERIA

B.Sc. DEGREE EXAMINATION

FACULTY: COMPUTING & APPLIED SCIENCES
DEPARTMENT: MATHEMATICS
SEMESTER: PRIMO
COURSE CODE: STA 111
COURSE TITLE: DESCRIPTIVESTATISTICS
INSTRUCTION: ANSWER ANY FOUR (4) QUESTIONS

SESSION: 2023/2024
CREDIT UNIT: 3

TIME: 3 HOURS

1. a. What are index numbers? (5marks)
   b. Outline any four uses of index numbers. (4marks)
   c. The table below gives the details of price and consumption of four commodities for 2010 and 2012. Using an appropriate formula, calculate an index number for 2012 prices with 2010 as base year. (8.5marks)

Commodities	Price per unit 2010 (₦)	Price per unit 2012 (₦)	Consumption value 2010 (₦)
Yam flour	70	85	1400
Vegetable oil	45	50	720
Beans	90	110	900
Beef	100	125	600

[17.5marks]

2. The following table shows the distribution of the masses of 120 logs of wood, correct to the nearest kg.

Mass/kg	Frequency
15-24	4
25-34	35
35-44	49
45-54	24
55-64	6
65-74	2

a) Draw a histogram for the distribution. (7marks)
b) Draw a cumulative frequency curve and use it to estimate, correct to two significant figures, the upper quartile. (10.5marks)

[17.5marks]

3. The following table shows the distribution of masses of 100 blood donors in Topfaith University.

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Mass/kg	60-62	63-65	66-68	69-71	72-74
No. of donors	5	18	42	27	8

Using the assumed mean of 67kg, calculate:

i) the mean mass (7marks)

ii) the standard deviation of the distribution. (10.5marks)

[17.5marks]

4. a) What is skewness? (4marks)

b) Write briefly on the following:

i) Normal distribution (3marks)

ii) Negatively skewed distribution (3marks)

iii) Positively skewed distribution (3marks)

c) What is Kurtosis? (4.5marks)

[17.5marks]

5. The table below shows the distribution of examination marks of 73 students of the Department of Mathematics in Topfaith University offering Business Statistics

Class interval	frequency
10-19	1
20-29	4
30-39	7
40-49	25
50-59	19
60-69	15
70-79	2

Calculate the Pearson coefficient of skewness, k_p. **(17.5marks)**

(Outline your workings clearly.)

6. The following are the length (in cm) of fifty planks cut by a machine in Topfaith wood processing factory.

33	49	81	58	59	71	42	88	68	91
54	32	50	59	41	55	38	56	86	62
50	69	23	84	77	33	71	42	69	93
61	51	46	76	63	96	26	70	66	80
42	52	60	33	68	39	61	71	48	66

a) Using class intervals of 21-30, 31-40, etc., construct the frequency table. (5marks)

b) Identify the median class. (2.5marks)

c) Estimate the median of the distribution. (5marks)

d) Estimate the mode. (5marks)

[17.5marks]

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MARKING SCHEME

Question No. 1

a. Statistics is the scientific method of collecting, organizing, summarizing, presenting and analyzing data as well as making valid conclusion based on the analysis carried out. It is a scientific method which constitutes a useful and quite often indispensable tool for the natural, applied and social workers. The method of statistics are useful in an ever-widening range of human endeavour, infact, in any field of thought in which numerical data exist. **5mks**

b. **U**ses of Statistics

i) It is used in decision making 1mk
ii) It is used in the analysis of yield of crops 1mk
iii) It can be used to measure an up-to-date expenditure pattern, revenue and estimates. 1mk
iv) It can be used to aid findings in astrology, chemistry, geology, etc. 1mk

c.

Commodities	Price per unit 2010 (₦) p_o	Price per unit 2012 (₦) p_1	Consumption value 2010 (₦) p_oq_o	2010 Qty q_o	p_1q_o
Yam flour	70	85	1400	20	1700

Vegetable oil	45	50	720	16	800
Beans	90	110	900	10	1100
Beef	100	125	600	6	750
			$\sum p_o q_o =$ 3620		$\sum p_1 q_o =$ 4350

4.5mks

Therefore, the Laspeyre's price index for 2012 with respect to base year 2010 is given by

$$P_{01}^{La} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 \qquad 2mks$$

$$= \frac{4350}{3620} \times 100 \qquad 1mks$$

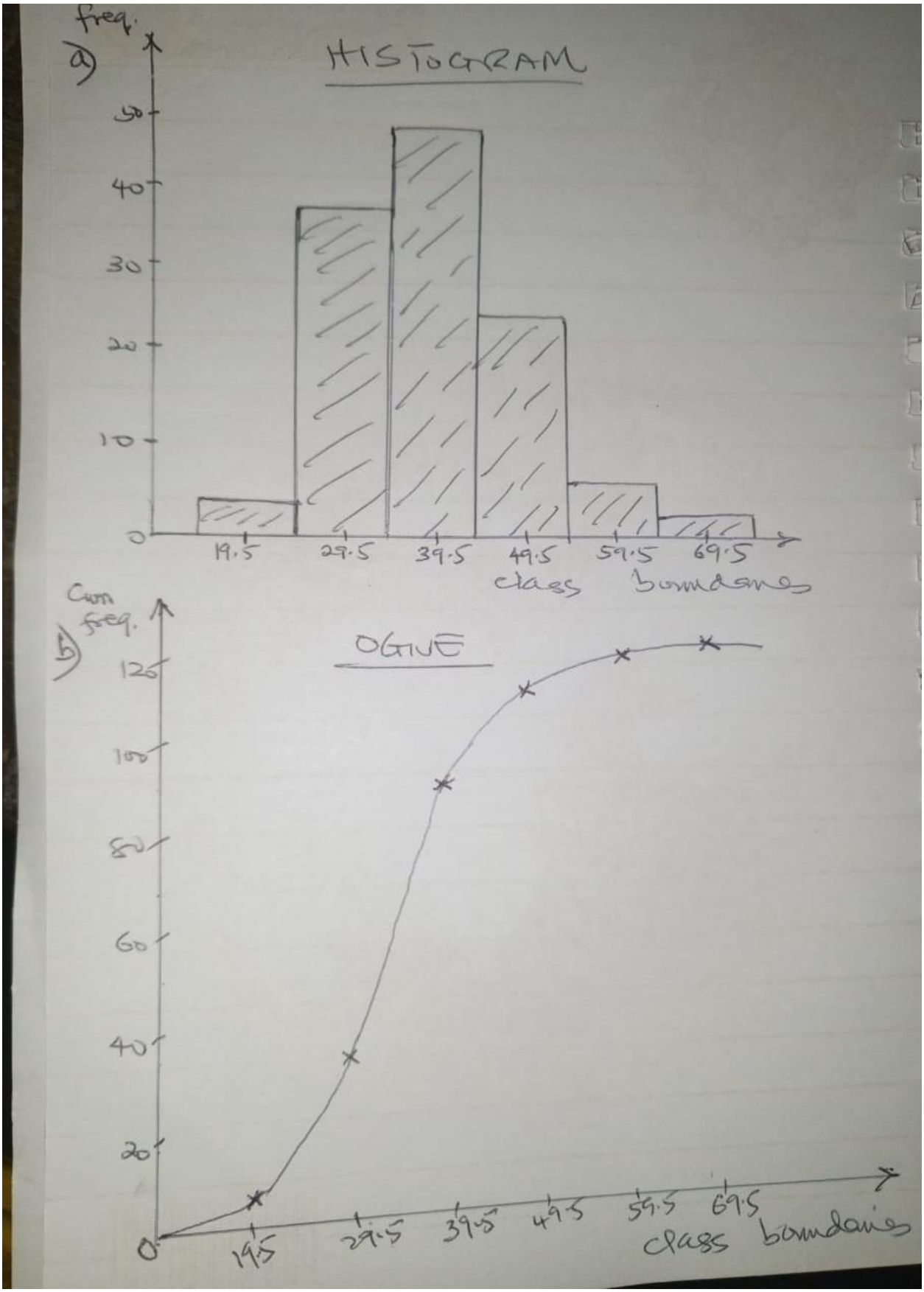
$$\therefore P_{01}^{La} = 120.7 \qquad 1mks$$

[Total = 17.5marks]

Question No. 2

Mass/kg	frequency	Class boundaries	Cum. freq.
15-24	4	14.5-24.5	4
25-34	35	24.5-34.5	39
35-44	49	34.5-44.5	88
45-54	24	44.5-54.5	112
55-64	6	54.5-64.5	118
65-74	2	64.5-74.5	120
	$\sum f = 120$		

7.5mks


a)
freq.
HISTOGRAM
50
40
30
20
10
0
19.5
29.5
39.5
49.5
59.5
69.5
class boundaries
b)
Cum freq.
OGIVE
120
100
80
60
40
20
0
19.5
29.5
39.5
49.5
59.5
69.5
class boundaries

a) **4mks** b) **4mks**

Now for the Upper quartile Q_3, $Q_3 = \frac{3(N+!)}{4} = \frac{3(121)}{4} = 90.75$ $1mk$

$\therefore Q_3 = 38.5$ **1mk** **[Total 17.5marks]**

Question No. 3

Mass/kg	Class mark (x)	f	d = x-A	u = (x-A)/c	fu	fu^2
60-62	61	5	-6	-2	-10	26
63-65	64	18	-3	-1	-18	18
66-68	67	42	0	0	0	0
69-71	70	27	3	1	27	27
72-74	73	8	6	2	16	32
		$\sum f =$ 100				$\sum fu^2 = 97$

5mks

i) $\bar{x} = A + c\frac{\sum fu}{\sum f}$ $2mks$

$\bar{x} = 67 + 3 \times \frac{15}{100}$ $2mks$

$\bar{x} = 67 + 0.45$ $1mk$

$\therefore \bar{x} = 67.45$ $1mk$

ii) $S.D = c \times \sqrt{\frac{\sum fu^2}{\sum f} - \left(\sqrt{\frac{\sum fu}{\sum f}}\right)^2}$ $2mks$

$S.D = 3 \times \sqrt{\frac{97}{100} - \left(\sqrt{\frac{15}{100}}\right)^2}$ $1mk$

$S.D = 3 \times \sqrt{0.97 - 0.0225}$ $1mk$

$S.D = 3 \times 0.9734$

$\therefore S.D = 2.92kg$ 2.5*mks*

[Total=17 ½ marks]

Question No. 4

a) **Skewness** can be defined as the measure of the degree of the deviation of a statistical distribution from the symmetry. **4mks**

b) i) **Nominal Scale:** This represents he most unrestricted assignment of numerals, infact, the numerals is used only as labels and thus words or letters would serve as well. The nominal scale has no direction and is applicable to qnumerals or letters derived from qualitative data. It is merely a classification item and has no other properties. **3mks**

ii) **Ordinal Scale:** This has magnitude, hence, it is a step more developed than the nominal scale. It has the structure of order preserving group. Here, items are placed in order of magnitude. Infact, it a group which includes transformation by all monotonic increasing function. **3mks**

iii) **Interval Scale:** This is used to specify the magnitude of observations or items. It is a higher scale of measurement superior to both nominal an ordinal scales Thus, it incorporates all the properties of both nominal and ordinal scales, and in addition requires that the distance between the classes be equal. **3mks**

c) **Kurtosis** is the measure of the degree of peakedness or flatness of a distribution. It can be either platykurtic, mesokurtic or lyptokurtic. **4.5mks**

[Total=17 ½ marks]

Question No. 5

Class	f	x	fx	Class boundaries	Cum. Freq	$\lvert x-\bar{x}\rvert^2$	$f\lvert x-\bar{x}\rvert^2$
10-19	1	14.5	14.5	9.5-19.5	1	1232.01	1232.01

20-29	4	24.5	98	19.5-29.5	5	630.01	2520.09
30-39	7	34.5	241.5	29.5-39.5	12	228.01	1596.07
40-49	25	44.5	1112.5	39.5-49.5	37	26.01	650.25
50-59	19	54.5	1035.5	49.5-59.5	56	24.01	456.19
60-69	15	64.5	967.5	59.5-69.5	71	222.01	3330.15
70-79	2	74.5	149	69.5-79.5	73	620.01	1240.02
	$\sum f =$ 73		$\sum fx =$ 3618.5				$\sum f\|x - \bar{x}\|^2 =$ 11024.73

5mks

Median, $Q_2 = l_1 + \left(\frac{\frac{N}{2} - \sum f_{med}}{f_{med}}\right)c$ **1mks**

$$Q_2 = 29.5 + \left(\frac{\frac{73}{2} - 12}{7}\right)10 \qquad 1mks$$

$$= 29.5 + 35$$

$$\therefore Q_2 = 64.5 \qquad 1mk$$

Mean, $\bar{x} = \frac{\sum fx}{\sum f}$ $1mks$

$$\bar{x} = \frac{3618.5}{73} \qquad 1mk$$

$$\therefore \bar{x} = 49.6 \qquad 1mk$$

Standard Deviation, $\sigma = \sqrt{\frac{\sum f|x - \bar{x}|^2}{\sum f}}$ $1mks$

$$\sigma = \sqrt{\frac{11024.73}{151.02}} \qquad 1mk$$

$$\therefore \sigma = 12.3 \qquad 1mk$$

Therefore, the Pearson Coefficient of Skewness,

$$k_p = \frac{3(\bar{x} - Q_2)}{\sigma} \qquad 1mk$$

$$k_p = \frac{3(49.6 - 64.5)}{12.3} \qquad 1mk$$
$$\therefore k_p = -3.6 \qquad 1mk$$

This means that the distribution is negatively skewed. **1/2mk**

[Total = 17.5marks]

Question No. 6

a) **Frequency table**

Length(cm)	freq (f)	x	fx	Cum. freq	Class boundaries
21-30	2	25.5	51	2	20.5-30.5
31-40	6	35.5	213	8	30.5-40.5
41-50	9	45.5	413.1	17	40.5-50.5
51-60	9	55.5	499.5	26	50.5-60.5
61-70	11	65.5	720.5	37	60.5-70.5
71-80	6	75.5	453	43	70.5-80.5
81-90	4	85.5	342	47	80.5-90.5
91-100	3	95.5	286.5	50	90.5-100.5

5mks

For the median: $\frac{N+1}{2} = \frac{50+1}{2} = 25.5 \qquad 1mk$

b) Hence, the median class is 51-60 **1.5mks**

c) Median, $Q_2 = l_1 + \left(\frac{\frac{N}{2} - \sum f_{med}}{f_{med}}\right)c \qquad 2ks$

$$= 50.5 + \left(\frac{\frac{50}{2} - 17}{9}\right)10 \qquad 2mks$$
$$= 50.5 + 8.89 \qquad 1mk$$
$$\therefore Q_2 = 59.4cm \qquad 1mk$$

d) $Mode = l_1 + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right)c \qquad 1mk$

$$Mode = 60.5 + \left(\frac{11-9}{(11-9)+(11-6)}\right)10 \qquad 1mk$$

$$= 60.5 + \left(\frac{2}{2+5}\right)10 \qquad 1mk$$

$$= 60.5 + 2.85 \qquad 1mk$$

$$\therefore Mode = 63.35 \qquad 1/2mk$$

[Total = 17.5marks]